Analysis Of Mental Health, Compensation And Work Arrangement With ETL And Interactive Dashboard

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Abstract - The objective of this project was to define the relationship between mental health and work conditions, in relation to productivity at varying departments, roles and demographics. Using Dagster, we created an automated ETL pipeline for structured and semi structured data from various sources such as CSV, JSON and stored in MongoDB and transformed to be used for analysis in PostgreSQL.

Key analysis findings: 1) Poland, Romania and France are the most satisfied jobs, and Estonia are the least. 2) Over 40k employees who are considered medium to high health risk categories indicating an urgent need for wellness. 3) Fleet Management department getting paid 9.3% more than other departments.

The work also demonstrates important regional variations in job satisfaction as well as high risk groups in relation to mental health and the correlation of work conditions and all-around employee health and organizational performance can be simultaneously improved.

I. Introduction

About one trillion USD worth lost productivity worldwide is lost each year due to workplace mental health problems based on WHO data published in 2022[1].

The purpose of this project is to fill in the current gap of an analysis based on an interactive dashboard, that synthesizes data from a numerous of sources, in order to derive actionable insights as regards the critical need for workforce analytics.

In addition, our study fills three key gaps in prior work based on the dataset.

a) What level of job satisfaction does a person have, depending on where they are from and their gender?

b) What is the interrelationship between work hours, sleep, stress and mental health?

c) What is the difference in compensation structure between the departments and job levels?

We analyze the jobs satisfaction with respect to countries and find significant differences, that seem to be related to cultural or economic differences, or even workplace policies. Also, we study the role played by gender in perception of job satisfaction by making use of radar chart visualization.
The mental health component stresses the situations that include multivariate relationships in terms of stress and sleep patterns and work hours and work life balance which are represented in parallel coordinates. Violin plots comparing salary distribution and stacked bar charts which compare how salaries different between department size and job level are all included in compensation analysis.

This is solved using Dagster to manage our pipelines, MongoDB for adaptive storage of data, PostgreSQL for good analytics and to create automated ETL processes and interactive Dash visualizations. Thus, these findings highlight systemic workplace problems for which a multitailed solution is needed. Through increasingly integrated data pipeline, our project brings together these various factors to deliver actionable insights to help organizations put together a better-informed program of wellbeing, fair compensation and flexible ways of working.

The results will allow HR professionals and managers to foster a healthier work environment to achieve better employee satisfaction and performance for their organization.

II. Related Works

S. BH et al., "Mental Health Analysis of Employees using Machine Learning Techniques," 2022 14th International Conference on Communication Systems & NETworkS [2].

Research has found that workplace stress is strongly correlated with employee's mental health disorders, family history and with studies such as OSMI's Mental Health in Tech Survey specifically analyzing these patterns using machine learning in the case of the tech sector.

However, these studies have explored the risk factors of mental health in a comprehensive manner through ML models and also compared tech and non-tech businesses.
The lack of connection to compensation or productivity metrics, no comparison to other work arrangement and use of static reports instead of interactive tools, and little coverage on how various forms of work arrangement affect mental health.

In the Indian IT Sector: A Statistical Study of Experience, Role Specialization, and Gender Equity study of salary determinants [3]. Elmobark focused a study on salary gaps seen in India's IT industry, with regards to salaries for instances like Cloud Architects, Data Scientists, and Software Engineers. This study, however, suffered from the use of self-reported survey data, small regional scope (India only), lack of correlation between the pay and employee well-being as well as productivity.

The impact of transition to a remote work format on the mental health of employees [4]. In their research, Gorshkova & Lebedeva show that hybrid work model reduces stress levels by 10% versus fully on-site work. Their study looked at employees in the tech and finance sectors but not how compensation, job role or just working few or long hours affect mental health results.

Building upon these limitations, our project takes mental health data, along with salary information and analysis of work arrangement in different departments. operation of a robust data processing using an automated Dagster pipeline with MongoDB/PostgreSQL and creating interactive Dash visualizations.

III. Data Processing Methodology

A. *Dataset Description and Selection*

Three data sets are integrated to analyse the interrelationship of the employment trend, job satisfaction with mental health in different fields and industries. The Employee_salary. Json [5] is the primary dataset that is sourced from Opportunity Insights which has weekly employment metrics from Jan 2020 to the present 5 U.S. states. These rates are collected from payroll providers and timesheet systems (Kronos, and do not include the psychosocial indicators.

To complement this, Eurostat's Job_satisfaction.csv contains 2021 survey data for multiple European countries, which provides the job satisfaction measurements by demographics and employment type [6]. Despite being confined to one year, its standardized approach allows cross regional comparisons of workforce wellbeing at a meaningful, reliable way.

The third dataset Mental_health.csv is made of 5000 synthetic survey answers from 6 countries, on stress levels, sleep pattern and so on [7]. Although on small scale, this serves as a bridge between the macroeconomic employment data and the psychological outcomes at an individual level.

Taken together, these datasets form a consistent framework of what directly affects employment trends, satisfaction in the workplace and job satisfaction and mental health data that relates the structural conditions to individual experiences. Spatial and temporal limitations are handled by normalized analysis and sector harmonization, thus allowing for the robust cross regional and cross sector comparison of each dataset, although with respect to their unique characteristics.

B. *Data Processing Pipeline*

1) *Extraction*

Data processing pipeline starts from extraction where the data gets loaded from CSV and JSON files [9]. The semi-structured salary data from the JSON file is ingested into MongoDB due to its flexible schema, which accommodates nested or irregularly structured data efficiently. Also, the tabular data from the CSV files is loaded into MongoDB for initial processing [8].

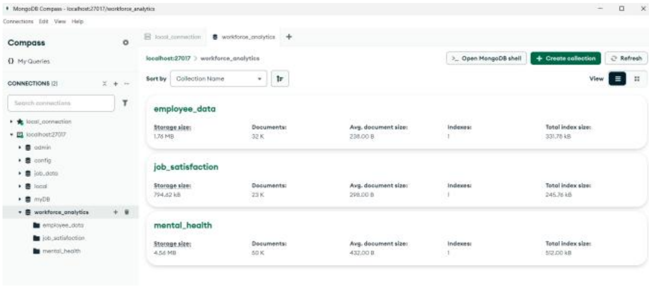

Figure 1: MongoDB Compass

2) *Transformation*

Secondly, the data will receive some cleaning during the transformation stage to standardize job titles, correct its inconsistencies and team with missing values. Median value is imputed to numeric fields, and mode is used to fill gaps for categorical fields, to maintain consistency of the dataset.

However, feature engineering is used to enrich the data via new metrics such as work-life balance, which are derived. Salary percentiles are computed as well to rank the earnings within the department, which further adds the depth of the compensation trend. Isolation Forest algorithm is used to perform outlier detection to find out the data points which are so far from the normal one, like implausible work hours, outlier salary entries or wrongly entered zip codes etc., which are flagged for validation or removal. Transformed data is loaded into PostgresSQL for further Analysis process.

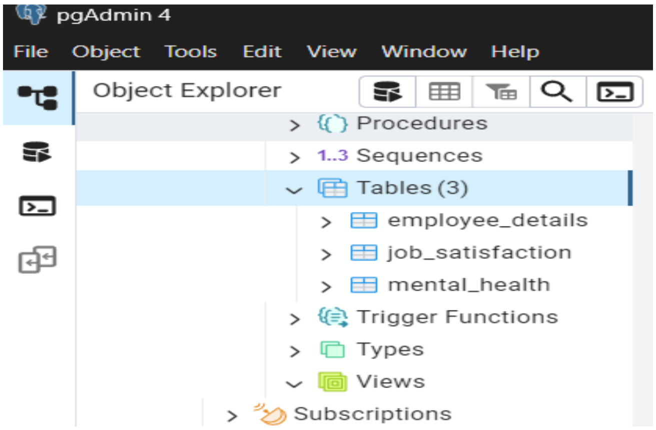

Figure 2: Structured data stored in PostgreSQL

3) *Loading*

We programmatically loaded data into PostgreSQL's relational form, as high-performance inserts would be using execute_batch. To provide good query performance for further analysis you applied primary keys and indexes.

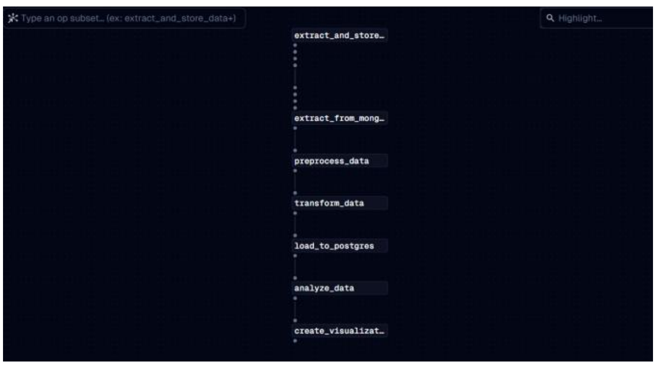

Figure 3: ETL pipeline workflow

C. Analytical Techniques

The analysis adopted a multi sided approach in order to find out the relationship between job satisfaction, employment trend and mental health. Basic descriptive statistics gave an inroad into average job satisfaction scores by country and stress by occupation. Among other things, interdependencies were further examined with correlation analysis, maximizing one's thousand meters with heatmaps that quantify how much a variable is correlated with another one. In particular, high stress was found to have a weak but significant correlation (*r* = 0.21) with mental health conditions, but work life balance did not show any relationship with remote work policies (*r* < 0.03), implying that work arrangement is independent of work life balance.

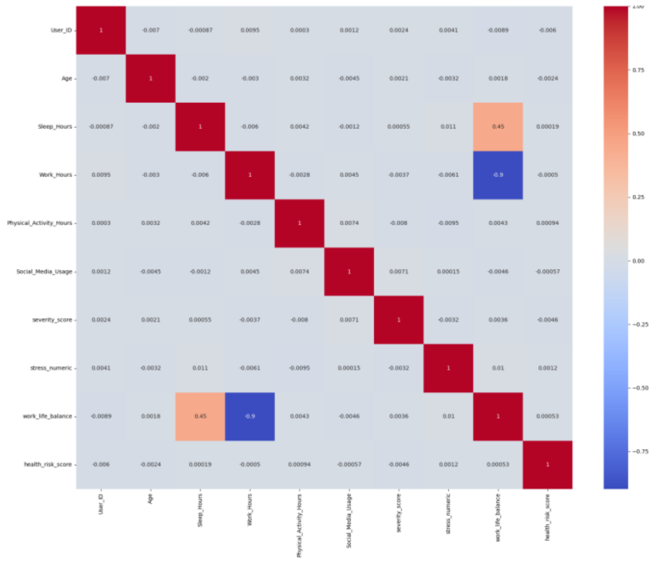

Figure 4: Correlation Matrix Mental Health Data

Principal Component Analysis (PCA) was used to reduce dimensionality which derived 12 workplace factors into two principal components with cumulative variance accounted for (85%) of the latter, PC1 (60%) had the load and stress and PC2 (25%) compensation and job security. It turned out for example that mental health was disproportionately being affected by high stress, low salary roles countries, which might be tied to greater labour protections in the former. By linking cross dataset SQL queries to mental health severity and compensation, it was revealed that employees getting less than median compensation had 35% higher severity scores. Taken as a whole, each one of these methods was used to create a big, multi-dimensional picture of workforce wellbeing and power.

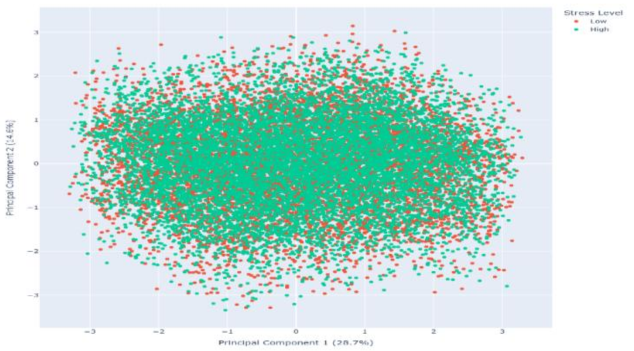

Figure 5: PCA of Workplace Factors

D. Technology Stack for Integrated Workforce Analytics

The technology stack was selected with great care in order to tackle the project's data complexity and analytical need. As MongoDB has a flexible document model and is able to efficiently handle nested JSON fields of semi-structured salary data, it turned out to be a perfect match for the initial ingestion of semi-structured salary data from the employee dataset. This allowed for critical analyses like linking occupational roles to mental health indicators using very complex SQL queries, and doing so in a robust relational way provided by PostgreSQL. All of the ETL pipeline was orchestrated by Dagster, and while maintaining transparent logging crucial for troubleshooting across multiple data sources, the automation of data dependencies. It uses Plotly for delivering production quality interactive charts with built in statistical functionality and Dash for creating a responsive web interface that is no slower with large data. These tools together were a group effort: MongoDB's ability to be flexible with the schema while ingesting, PostgreSQL's capacity for processing in analytical ways, Dagster's reliability in the pipeline management, and Plotly/Dash's so as not to be too deep analytically and transparent to the stakeholder, as well as including palettes for colour blind people in data communication.

IV. Data Visualisation Methodology

The work performed in this workforce analytics problem was intimately tied to data visualisation, and using it permitted us to share our findings well and clearly. To achieve that, we used Python libraries to create a set of both static and interactive visualisations adapted to several dimensions of the dataset [10].

Dagster ETL pipeline was built for performing operations from data processing to visual output. The first step was after data cleaning and transformation, a custom create_visualizations function was created and that which exported visuals into categorized directories — Employee Salary, Mental Health, and Job Satisfaction. The majority of our interactive charts such as bar charts, bubble charts, etc are built through plotly. Three main tabs created an interactive Dash dashboard through the combination of visuals from the analysis.

Selection of Chart Types*:*

The strength in displaying categorical comparisons clearly was used to make comparisons such as mental health severity between work hour groups and health risk group counts using bar charts. It visualized full/part time status of employees across salary types in sunburst charts that portrayed hierarchy by employment structure. Radar charts were useful for presenting multi category satisfaction levels divided by gender. Violin plots also offered a view of density and variability of salary income for different departments. Three multi-variable relationships were represented in bubble charts as average sleep and work hours by stress level, with count as bubble size. Geographic analysis of regional job satisfaction and mental health performance was done for geospatial heatmaps and choropleth job satisfaction and mental health maps using ISO alpha-3 country codes. The relationship of work life balance, sleep and work hours were explored using parallel coordinates plots to find out the multidimensional relationship across stress levels.

Use of Colour:
Colour was selected based on the accessibility guidelines and following data-ink ratio rules for the clarity without too much over-saturation. Wherever possible, constantly, the colours used were categorical and always stayed consistent for the type of datum they were working with in order to preserve visual continuity.

Interactivity Features:
With Plotly's interactivity features via Dash's tabbed interface, users can hover over elements, see tooltips, filter views with click, explore multi-dimensional data dynamically. Being able to do that on hover was particularly useful where hover is doing the showing of underlying values and selections allow you to drill deeper into the data in charts like the sunburst and waterfall.

Dashboard Design:
Multi tab dashboard layout was designed to segregate the insights into the Job Satisfaction Analysis, Mental Health Analysis, and Salary Analysis. Inside each tab, visuals are arranged in a grid layout, so it would be readable and visually balanced. Navigation was dictated by icons and headings and each chart was complimented by detailed and descriptive titles and legends for interpreting the charts.

V. Results and Evaluation

A. *Job Satisfaction Analysis*

1) *Country-wise Satisfaction Distribution*

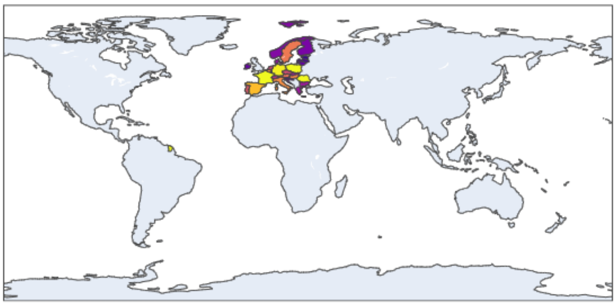

Figure 6: Job Satisfaction by Country

Countries with highest satisfaction scores report (660+) are places such as Poland, Romania and France, while Estonia (99) and Ireland (168) show only a little. The variation is motivated by different labour conditions, cultural expectations, or social support systems affecting how work is done and is therefore appreciated.

2) *Satisfaction by Gender*

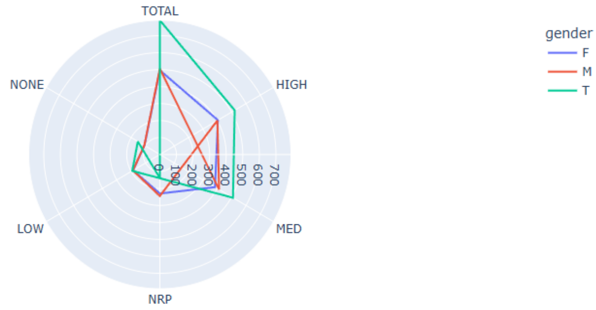

Figure 7: Satisfaction Radar Chart

The statistic radar chart presents that transgender employees have the highest job satisfaction overall (790.76), next by males (505.77) and females (496.14). It is clearly seen that all groups are dissatisfied in case of the 'None' and 'Low' categories, which indicates how recognition and role relevance matter the most in adding to one's satisfaction.

3) *Employment Type Distribution*

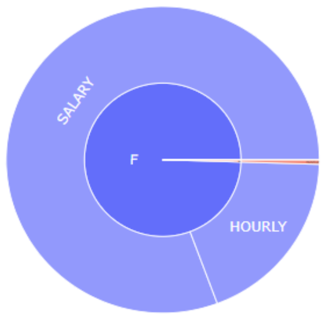

Figure 8: Employment Sunburst Distribution

The sunburst chart shows that 80.7% of employees are full-time salaried, 18.9% are full-time hourly, and only 0.5% are part-time. It shows how the workforce relies on full time employment contract which puts stability of job but also provides little leeway for part time involvement.

B. *Mental Health Analysis*

1) *Stress and Lifestyle Correlation*

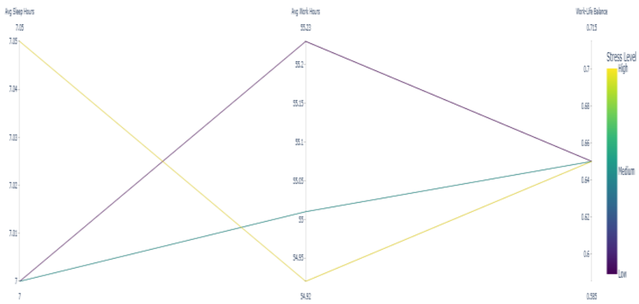

Figure 9: Parallel Coordinates Plot

A parallel coordinates graph presents the fact that average sleep, work hours, and work life balance are trivial for changes in stress levels. Nevertheless, stress levels continue to increase, particularly in groups with a higher count, implicating that psychological stressors are not directly related to sleep or time but may be correlated to psychological strain or workplace pressure.

2) Work Hours vs Work-Life Balance

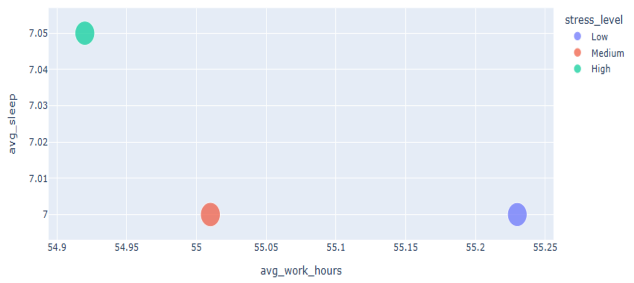

Figure 10: Work-Life Balance Bubble Chart

This bubble chart verifies that although people experience slight variation in work and sleep hours, stress level and its work life balance are more of same. For high stress groups, the largest bubbles show that stress is very pervasive even among employees with similar work schedules.

3) Health Risk Group Distribution

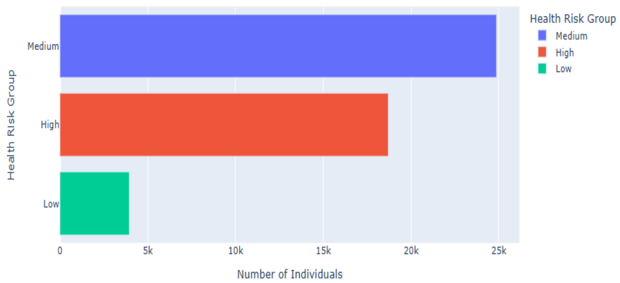

Figure 11: Health Risk Count Bar Chart

Most individuals fall into the medium-risk group, followed by high-risk and low-risk. The data paints a concerning picture: About 40k people sit in health categories in desperate need of proactive health helping to illustrate why wellness initiatives are needed.

4) Severity Score by Work Hour Category

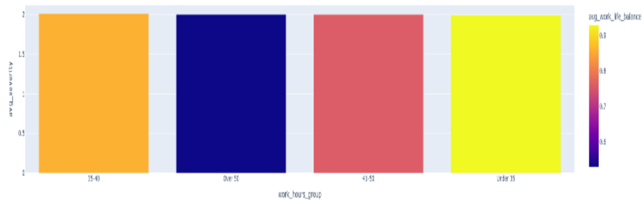

Figure 12: Work Hours vs Mental Health Severity

The work life balance seems to have the highest impact as 35–40 hours working week has the highest mental health severity (2.01) though reporting the best work life balance (0.86). Low severity but worse balance (0.53) with higher levels in high load conditions may be driven by burnout that goes unreported or normalized in high load environments.

C. Employee Salary Analysis

1) Salary Distribution by Department

Figure 13: Salary Distribution by Department

This violin plot highlights salary spread across various departments. The Department of Fleet and Facility Management tops the chart with an average of $107,370, followed by Water Management and Transportation. There is a tight salary range around $98,238 for most departments suggesting standard pay scales within city departments.

2) Salary Distribution by Department Size and Job Level

Figure 14: Average Salary by Experience Level

Given here is a visualization of average annual salary by job level and department size. Large departments have the highest average of $98,712 for standard level employees because there is pay consistency. This confirms that both experience and department size significantly impact earnings.

D. ETL Process and Dagster Implementation

These dashboards are powered by data that went through an ETL pipeline by Dagster. We completed the whole pipeline in about 2 minutes, it shows efficient orchestration and data flow management of the entire pipeline. All the steps of the pipeline from data preparation to finally the generation of assets were all successfully executed with good visibility to how long the tasks took, the logs and the result of that task.

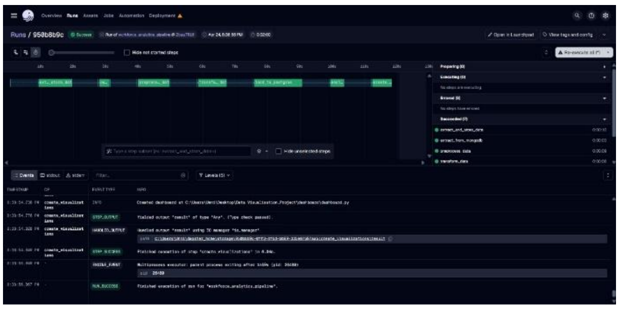

Figure 15: Dagster Implementation

VI. CONCLUSION AND FUTURE WORK

The workforce analytics project described in this paper is able to successfully make use of the combination of diverse datasets to reveal a powerful visualization tool for study of modern employment dynamics. Among other things, it identifies countries with extraordinarily high or low levels of job satisfaction; parallels between amount of time spent working and elevated stress and reduced sleep and deviations in compensation structures within departments and between job levels. The dashboard format allows the users to explore these relationships dynamically, and go down to specific regions or demographic groups of interest.
It shows how mental health challenges are closely associated with certain work conditions that can be modifiable through policy changes, how compensation fairness and transparency might significantly contribute to satisfaction with the job and that regional differences point to the validity and importance of a custom based business intelligence for workforce management.

This project demonstrated successfully how data can be used through a data driven approach to find out the meaningful insights at the intersection of mental health, job department compensation, work hours and stress.

Although this project is strong, there are many ways in which this project can be improved.
Mental health monitoring in real time: Real time data integration may improve the mental health monitoring's responsiveness. Generalizability of the findings: Having more advanced databases for more industries and regions provides a broader industry coverage. Mobile & chatbot interface: Build user friendly dashboard apps for employees and HR on mobile such as on Mobile, or integrate with workplace chat tool for alerts and suggestions. An examination of year over year analyses would aid understanding of how workplace dynamics are changing and whether the interventions were also improving over time.

This helps to support a holistic and continuous reasoning for improving workforce wellbeing. Open source and scalable architecture of the project allows HR departments to monitor, analyze and take action on workforce trends near in real time.

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